# Workshop 2

## Members

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## Issue 1

For inputs like "tomorrow 11:00, I prefer regions SG", the agent called `recommend_best` with a guessed ISO (e.g. `2023-11-10T11:00:00`), causing "No points in window" even when data existed.

## Solution for Issue 1

- Expose a single tool `recommend_best_from_text(text_time)` that accepts natural-language time (e.g. `"tomorrow 11:00"`).
- The tool internally parses time and runs the recommendation; the LLM never passes raw ISO strings for scheduling.
- System prompt instructs: use `recommend_best_from_text` with the user's time text; do not invent dates or ISO.

## Outcome after solution

```
You: I want to schedule a job tomorrow 10am, I prefer regions SG

[DEBUG] Messages this turn:
01. SystemMessage: '\nYou are a single-agent carbon-aware scheduler.\n\nGoal:\nRecommend the best (region, start time) aroun'
02. HumanMessage: 'I want to schedule a job tomorrow 10am'
03. AIMessage: ''
    └── tool_call: recommend_best_from_text({'text_time': 'tomorrow 10am'})
04. ToolMessage: '{"ok": true, "region": "SG", "start_time_sg": "2026-03-11T09:00:00+08:00", "g": 290, "desired_time_s'
    └── tool_result from recommend_best_from_text: {"ok": true, "region": "SG", "start_time_sg": "2026-03-11T09:00:00+08:00", "g": 290, "desired_time_sg": "2026-03-11T
05. AIMessage: 'I recommend scheduling your job in the SG region at 09:00 SG on 2026-03-11. The carbon intensity is '
06. HumanMessage: 'I want to schedule a job tomorrow 10am, I prefer regions SG'
07. AIMessage: ''
    └── tool_call: update_prefs({'regions_allowed': 'SG'})
    └── tool_call: recommend_best_from_text({'text_time': 'tomorrow 10am'})
08. ToolMessage: '{"regions_allowed": ["SG"], "allowed_shift_minutes": 60}'
    └── tool_result from update_prefs: {"regions_allowed": ["SG"], "allowed_shift_minutes": 60}
09. ToolMessage: '{"ok": true, "region": "SG", "start_time_sg": "2026-03-11T09:00:00+08:00", "g": 290, "desired_time_s'
    └── tool_result from recommend_best_from_text: {"ok": true, "region": "SG", "start_time_sg": "2026-03-11T09:00:00+08:00", "g": 290, "desired_time_sg": "2026-03-11T
10. AIMessage: 'I recommend scheduling your job in the SG region at 09:00 SG on 2026-03-11. The carbon intensity is '

Agent: I recommend scheduling your job in the SG region at 09:00 SG on 2026-03-11. The carbon intensity is 290g, shifted by -60 minutes.
```

## Issue 2

When the user says "I want to schedule a job today, recommend me the best" (or "tomorrow" with no time), the parser treats the missing time as 00:00. The forecast data usually only has slots during the day (e.g. 09:00–12:00), so the ±shift window around midnight has no points and the agent replies "No points in window" instead of recommending the best slot on that day.

## Solution for issue 2

If the time string is exactly "today" or "tomorrow" (no time part), treat it as "best slot on that day": add `_recommend_best_for_day(day_key)` that finds the forecast point with minimum carbon intensity on that date across allowed regions, and call it from `recommend_best_from_text_tool` before parsing a single clock time. Date alignment to the mock forecast is reused so it works with static workshop data.

## Outcome after solution

```
You: I want to schedule a job today, recommend me the best

[DEBUG] Messages this turn:
01. SystemMessage: '\nYou are a single-agent carbon-aware scheduler.\n\nGoal:\nRecommend the best (region, start time) aroun'
02. HumanMessage: 'I want to schedule a job today, recommend me the best'
03. AIMessage: ''
    └── tool_call: get_profile({})
04. ToolMessage: '{"regions_allowed": ["US_WEST"], "allowed_shift_minutes": 60}'
    └── tool_result from get_profile: {"regions_allowed": ["US_WEST"], "allowed_shift_minutes": 60}
05. AIMessage: ''
    └── tool_call: recommend_best_from_text({'text_time': 'today'})
06. ToolMessage: '{"ok": true, "region": "US_WEST", "start_time_sg": "2026-03-11T09:00:00+08:00", "g": 220, "desired_t'
    └── tool_result from recommend_best_from_text: {"ok": true, "region": "US_WEST", "start_time_sg": "2026-03-11T09:00:00+08:00", "g": 220, "desired_time_sg": "2026-03
07. AIMessage: 'I recommend scheduling your job in the US_WEST region at 09:00 SG on 2026-03-11. The carbon intensit'

Agent: I recommend scheduling your job in the US_WEST region at 09:00 SG on 2026-03-11. The carbon intensity is 220g, shifted by 0 minutes.
```